Mohamed Atef Fahmi

Senior Data Engineer | Spark & Streaming | Data Platforms
Abu Dhabi, UAE | +971 507 422 363 | muhamedfahmy7474@gmail.com
LinkedIn | GitHub | Medium | Portfolio

Professional Summary

Senior Data Engineer with 6+ years of experience building and optimizing batch and real-time data platforms on Spark, Kafka, Airflow, Hadoop, and Databricks. Cut a Spark/Scala segmentation job from 16 hours to 2 hours (87.5%) at e& across tens of millions of records, and currently run 95+ production pipelines feeding national price indices at the Statistics Centre – Abu Dhabi. Strong in data quality automation, CI/CD, and pipeline observability, with hands-on work in ML infrastructure and LLM-assisted pipeline automation.

Professional Experience

Senior Data Engineer **Sep 2025 – Present**
Statistics Centre – Abu Dhabi (SCAD) *Abu Dhabi, UAE*

- Own 95+ daily production pipelines (Python, Scrapy, Playwright, Airflow) ingesting prices from 75+ websites and APIs into Presto/Trino and MySQL, powering the national Consumer Price Index (CPI) and retail price indices.
- Designed and built MARSAD solo in 3 weeks: a near-real-time pipeline monitoring platform (FastAPI, Vue.js, MySQL, Docker) deployed to OpenShift via GitHub Actions. It provides data quality checks, price outlier detection, and failure tracking for 130+ users across teams.
- Introduced pytest API-level testing and data contract validation, so pipelines meet requirements agreed with the price and income statistics teams before data reaches downstream consumers.
- Built an LLM-powered scraping agent (LangGraph, DSPy) that adapts scraping logic when website structures change, covering static and JavaScript-rendered pages and reducing manual pipeline repair.
- Set up ML infrastructure with MLflow for experiment and model tracking, and a synthetic data framework (SDV, CTGAN, TVAE, CopulaGAN) to enable privacy-conscious model development.
- Contribute to platform modernization on Databricks, Delta Lake, and Unity Catalog, translating requirements from statistics stakeholders into engineering solutions in Agile workflows.

Senior Data Engineer **Nov 2024 – Sep 2025**
e& (Etisalat) *Dubai, UAE*

- Re-engineered a Spark/Scala ETL framework behind customer segmentation, cutting runtime from 16 hours to 2 hours (87.5%) across tens of millions of customer records.
- Designed the data pipelines for a monetized, AI-powered customer segmentation platform used by external clients, including Careem and Amazon, to select audiences for targeted SMS and WhatsApp campaigns.
- Built real-time pipelines with Kafka, Kafka Streams, and Spark Streaming, with Redis in the real-time processing path.
- Optimized large batch workloads on HDFS (Spark, Hive, Impala, Presto), using partitioning, bucketing, and EXPLAIN-driven query tuning.
- Created a reusable ETL framework with dynamic Scala compilation, letting analysts define new pipeline logic without engineering support for each change.
- Used Databricks and Delta Lake for versioned ingestion and batch processing.

Data Engineer **Oct 2022 – Nov 2024**
LigaData *Remote*

- Built and maintained Hadoop/HDFS/Hive workflows, orchestrating daily, weekly, and monthly pipelines with Airflow and serving analytical queries through Presto/Trino.
- Cut report generation time by 40% through workflow automation and query optimization.
- Modeled analytical datasets using dimensional modeling and One Big Table patterns for downstream reporting.
- Deployed data workloads with Docker and Kubernetes for consistent execution across environments.

Data Engineer & Analyst - Contract **Jun 2022 – Dec 2023**
Upwork *Remote*

- Delivered ETL workflows and API integrations for e-commerce and logistics clients, and built ML models reaching up to 85% prediction accuracy on client use cases.

Data Analyst & Operations **Jun 2019 – Sep 2021**
Cube *Mansoura, Egypt*

- Built Excel and Power BI reports on website traffic and student performance to support operational decision-making.

Key Projects

Click-Streaming Data Pipeline GitHub

- End-to-end streaming pipeline (Kafka, Kafka Connect, Schema Registry, ksqlDB, Spark Streaming) that ingests, validates, enriches, and lands clickstream events in MinIO; fully Dockerized.

Kakashi – Open-Source PII Masking CLI GitHub

- Python CLI that masks PII in AI-agent workflows, supporting 50+ file formats and 20+ agents.

AWS Retail Data ETL Pipeline GitHub

- Batch ETL from MySQL on RDS to a star schema via AWS Glue, stored as partitioned Parquet in S3 and queried with Athena; infrastructure provisioned with Terraform.

Skills

Languages: Python, Scala, SQL, Bash

Processing & Streaming: Apache Spark, Spark Streaming, Kafka (Streams, Connect, ksqlDB), Hadoop/HDFS, Hive, Impala, Presto/Trino, Databricks, Delta Lake

Orchestration & Quality: Airflow, Data Contracts, pytest, Great Expectations, Dimensional Modeling

Databases & Storage: MySQL, PostgreSQL, SQL Server, Redis, S3, MinIO

Cloud & DevOps: AWS (S3, Glue, RDS, Athena), Azure (Data Factory, ADLS Gen2, Synapse), Docker, Kubernetes, OpenShift, GitHub Actions, Azure DevOps, Terraform

ML & AI Infrastructure: MLflow, SDV/CTGAN, LangGraph, DSPy, RAG

Ways of Working: Agile (Scrum/Kanban), Code Review, Technical Documentation

Education

Mansoura University – *B.Sc. in Computer Engineering*

Information Technology Institute (ITI) – *Postgraduate Diploma in AI & Machine Learning (EPITA, France), Grade A+*

Certifications

Microsoft Certified: Azure Data Engineer Associate (DP-203)

AWS Certified Cloud Practitioner

Udacity Data Engineering and Data Streaming Nanodegrees; Data Engineering ZoomCamp